

MOHAMMED MAKSOOD ALAM M

AI Engineer

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Summary

AI Engineer building production Applied Machine Learning and GenAI systems across RAG pipelines, Hybrid RAG, NLP, and multi-agent orchestration. Delivered an ed-tech tutoring chatbot, question-generation engine, automated grading system, and data engineering pipeline, collaborating with DevOps on deployment and CI/CD. Skilled in Python, Django, LangChain, Qdrant, and multi-provider LLM orchestration (OpenAI, Gemini, Anthropic).

Skills

- **AI & LLM** – Applied Machine Learning, NLP, RAG Pipelines, Hybrid Semantic Search, LLM Orchestration (LangChain), Agentic Frameworks (Google ADK, CrewAI), Multi-Agent Systems, Prompt Engineering, OpenAI / Gemini / Anthropic APIs, Ollama, Sentence Transformers, IndicBERT Embeddings
- **Languages** – Python, JavaScript, TypeScript
- **Backend & Infra** – Django, REST APIs, Node.js, Express.js, Celery, Redis, Kafka/Redpanda, Socket.IO, Docker, RunPod, AWS S3/CloudFront
- **Databases** – Qdrant, ChromaDB (vector), MongoDB, MySQL, SQL, Neo4j
- **Tools** – Git/GitHub, Gradio, OpenCV, PyMuPDF, Tesseract, Postman, Figma, Claude Code, Cursor
- **Soft Skills** – Collaboration, Communication, Problem-Solving, Adaptability, Leadership

Experience & Projects

AI Engineer | EdgeUp | May 2025 – Present

AI Tutoring Chatbot (RAG)

Tech: Python, Django, LangChain, Socket.IO, Kafka/Redpanda, Qdrant, Redis, Celery, OpenAI GPT-4o-mini, Gemini 2.5 Flash

- Built a curriculum-grounded RAG chatbot serving 500+ students with 4–7s response times, delivering context-aware answers from course material
- Architected a real-time messaging layer (Socket.IO + Kafka/Redpanda + Redis) supporting 1,000 concurrent users with sub-second latency
- Integrated multi-provider LLM failover (GPT-4o-mini to Gemini 2.5 Flash) for zero-downtime high availability across provider outages

Intelligent Question Generation Engine (Hybrid RAG)

Tech: Python, Django, Gemini 2.5 Pro, Qdrant, Neo4j, Redis, Sentence Transformers, IndicBERT

- Developed a Hybrid RAG engine (Gemini 2.5 Pro + Qdrant) generating thousands of MCQs across 22 courses and competitive exams (UPSC, TNPSC, SSC, Banking, RRB)
- Orchestrated a multi-agent validation pipeline achieving 90%+ acceptance rate and 30%+ LLM cost reduction via rate limiting and request deduplication

Automated Exam Grading & Analytics System

Tech: Python, Django, Celery Beat, Gemini, OpenCV, Sentence Transformers, AWS S3/CloudFront

- Developed an auto-grading pipeline (OpenCV OMR + Gemini) processing hundreds of exam scripts with 98%+ agreement with human graders
- Reduced grading time from hours to minutes and surfaced AI-driven student analytics, with Celery Beat scheduling and exponential-backoff retry logic for reliable result delivery

Multi-Course PDF Vectorization & Knowledge Ingestion Pipeline

Tech: Python, Django, Gemini 2.5 Flash, Qdrant, Neo4j, PyMuPDF, Tesseract OCR, Gradio, IndicBERT

- Designed a PDF-to-vector ingestion pipeline for 500+ documents across 13+ subjects, powering Neo4j knowledge graphs with multilingual IndicBERT embeddings
- Streamlined chunking with OCR fallback (PyMuPDF/Tesseract) and delivered a Gradio console for on-demand MCQ generation via Gemini 2.5 Flash

Junior Developer (Intern) | Zaryah Angels Pvt. Ltd. | Jan 2025 – Mar 2025

- Researched and evaluated AI coding assistants and agentic developer tools (Claude Code, Cursor, GitHub Copilot, Windsurf), assessing their capabilities for AI-assisted software development

Education

- Bachelor of Computer Applications (BCA) – The New College, Chennai | 2025 | GPA: 7.2/10
- HSC – Justice Basheer Ahmed Sayeed Memorial Matric. Hr. Sec. School | 2022 | 83.5%

Certifications: Android App Development – Techobytes / IIT Madras · MongoDB Atlas – MongoDB · SQL & Relational Databases – IBM